PILOT AI Factory Demo Specification (YAML + Sigma Comparison)

1. Objective

Demonstrate that digital actions can create physical consequences, and that falsified telemetry can conceal them, while PILOT detects causal inconsistency across domains. Include parameterised scenarios (YAML) and a correlational baseline (Sigma-style rules) for comparison.

2. System Architecture

- Digital layer: workload scheduler (GPU load per zone)
- Physical layer: simplified thermal model
- Telemetry layer: actual vs reported vs baseline
- PILOT layer: causal validation
- Correlation baseline: Sigma-style rules

3. Relevance Filter

Only data that defines or changes system state is retained. All other telemetry is discarded.

- Identity: roles, group membership
- Privilege: credential mappings
- Endpoint: execution context
- Network: connectivity constraints

4. Baseline Telemetry

Baseline telemetry defines expected system behaviour under normal conditions.

- 30% load → 36–38°C
- 60% load → 42–45°C
- 90% load → 48–54°C
- Neighbour zones increase slightly over time

5. YAML Scenario Template

```
scenario:
  name: asymmetric_gpu_load_with_falsified_telemetry
  duration_minutes: 20

digital:
  gpu_load:
    A: 0.92
    B: 0.35
    C: 0.30

physical:
  cooling_capacity:
```

```
    A: 0.60
    B: 0.60
    C: 0.60
  neighbour_heat_bleed: 0.08

telemetry:
  falsification:
    enabled: true
    zone: A
    reported_temp_offset_c: -9
    flatten_trend: true

baseline:
  expected_temp_at_load:
    "0.30": [36, 38]
    "0.60": [42, 45]
    "0.90": [48, 54]

pilot:
  checks:
    - thermal_inconsistency
    - cross_zone_inconsistency
    - reported_vs_inferred
```


6. Sigma-Style Correlation Rules (Baseline)

```
title: High Temperature Alert
logsource:
  product: ai_factory
detection:
  condition: temperature > 50
level: high

---
title: Load and Temperature Correlation
detection:
  condition: (gpu_load > 0.8) AND (temperature > 48)
level: medium

---
title: Zone Imbalance Detection
detection:
  condition: abs(temp_A - temp_B) > 10
level: medium
```

7. Scenarios

- Valid: load increase → temperature rise → both systems detect normal behaviour
- Operational issue: overheating → both systems detect issue
- Falsified telemetry: correlation misses, PILOT detects violation
- Incomplete data: PILOT returns uncertain

8. Key Distinction

Correlation evaluates signals. PILOT evaluates feasibility.

- Correlation: does this look suspicious?
- PILOT: could this have happened at all?

9. Output

- Valid
- Violation (causally impossible)
- Uncertain

10. Implementation Steps

- Simulate baseline telemetry
- Generate actual telemetry
- Inject falsified telemetry
- Apply relevance filter
- Run PILOT constraint checks
- Run Sigma rules in parallel
- Compare outputs

11. Bottom Line

The YAML defines the scenario, Sigma shows correlation limits, and PILOT proves causal validation.